CREATED BY RAPID FIRE

Performance OneROM v3

Fire board booklet

Connect your board, build and verify firmware, and choose the right emulator, ALDL or data trace mode.

Fire 28A / 28B / 28C + Fire 32A / 32B

Bundled One ROM firmware v0.8.0

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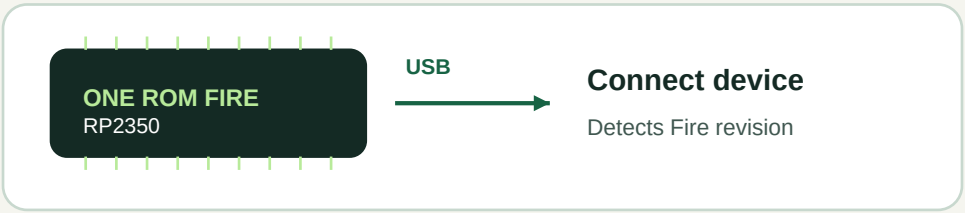
Before you start

Use the Windows x64 app, a USB data cable and a ROM file suited to your board and ECU setup. The build and programming workflow runs offline.

Connect and confirm the board



1. **Connect a Fire board**

   Open Performance OneROM v3. Plug the board in with a USB data cable, then select **Connect device**. Choose the intended Fire board if Windows shows a device picker.

2. **Confirm the board shown**

   When valid installed firmware identifies the board, the app fills in the Fire model and PCB revision. The build uses the bundled v0.8.0 firmware and applies the board pin settings for your chosen mode automatically.

3. **Check the board label if needed**

   If the app cannot identify the revision, check the physical board label and select the matching revision in **Custom image**. **Device information** shows what the app read from the board.

What you choose

Select your ROM image and the firmware mode that suits your serial setup. Check the suggested ROM type against your chip and image before building.

CUSTOM IMAGE WORKFLOW

Build, program and verify

ROM → Mode → Build → Program → Verify

1 Choose ROM image

In **Custom image**, select your original **.bin** or **.rom**. Check the suggested ROM type. Keep exact-size handling when the file already matches your chip.

2 Choose a system firmware ROM

Select a mode for the detected Fire family. Click **Build firmware**. Use **Save encrypted firmware** if you want an **.orfw** copy.

3 Program the Fire board

Click **Program device**, check the board and firmware shown, then choose **Erase and program**. This replaces the selected flash contents. Keep USB connected until verification finishes.

4 Wait for verification

The app reads the programmed image back and checks it byte for byte. Leave **Restart One ROM after programming** selected if you want the board to restart after the write.

OSE size example

A standard 32 KiB OSE image fits a Fire 28 / 27C256 configuration. A 128 KiB stacked OSE image uses a matching Fire 32 / 27C010 configuration.

Choose a firmware mode

The Fire 28 and Fire 32 families each offer the seven modes below. Select the entry for your board family. **DT** denotes data trace support.

NORMAL DT
Standard Ostrich emulation and a 38,400 baud UART bridge.

DISABLED DT
Standard Ostrich emulation and DT, with the auxiliary UART bridge disabled.

ALDL 808 DT
8,192 baud ALDL bridge with standard TX and RX byte polarity.

ALDL 808 TX Inv DT
8,192 baud ALDL. Inverts bytes sent out by the Fire board.

ALDL 808 RX Inv DT
8,192 baud ALDL. Inverts bytes received by the Fire board.

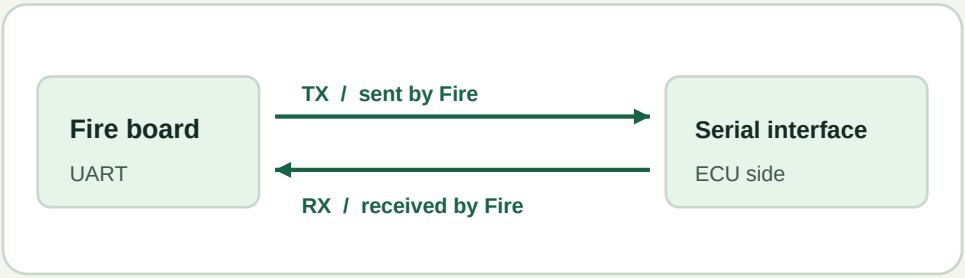
ALDL 808 TXRX Inv DT
8,192 baud ALDL. Inverts bytes in both directions.

OSTRICH Z DT
Ostrich Z protocol for compatible host software, with an HTS UART bridge at 38,400 baud.

No system firmware ROM

This separate menu choice builds an image without a system plugin. Select a DT variant when you need emulator or trace support.

Understand TX and RX



1. **Read direction from the Fire board**

   TX means transmit from the Fire board. RX means receive into the Fire board. The TX and RX names on an attached adapter are described from that adapter's point of view.

2. **Match the known interface setup**

   Choose standard, TX Inv, RX Inv or TXRX Inv using the mode already known to work with your interface. The inversion modes change the data bytes in the chosen direction. The firmware supplies the mode's UART settings.

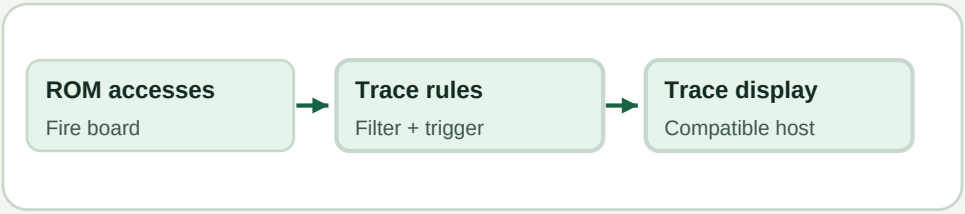
3. **Rebuild when changing the mode**

   Select the new mode, build firmware again and program the board. Confirm communication with the intended tuning software and interface.

Board detection and serial setup

USB board detection fills in the Fire board fields. The ECU and serial interface determine which ALDL polarity mode to select.

Use data tracing (DT)



1. **Build and program a DT variant**

   Select the right mode in **System firmware ROM**, build the image, then program and verify it. Reconnect to the running emulator from compatible tuning software.

2. **Set the trace in the host software**

   DT reports ROM addresses accessed by the ECU. A compatible host can request an address range, suppress consecutive duplicates, and set start or end triggers.

3. **Choose how addresses are reported**

   Supported trace options include absolute or relative addresses, continuous or packet-based reporting, packet limits and stop controls. The host must support the trace commands you select.

What you see

DT shows ROM address activity. ALDL is a separate serial data path for ECU diagnostics and logging.

REFERENCE

Saved firmware and troubleshooting

1 Reopen a saved image

Select **Firmware file**, then **Choose saved firmware**. The app unlocks a compatible **.orfw** automatically. Confirm the target board before programming.

2 Apply a new plugin or mode

A saved image keeps its original plugin. To apply the latest emulator fix or switch modes, build fresh firmware from the original ROM image and save a new **.orfw**.

3 Resolve a connection problem

Check the USB data cable and select the intended Fire device. If the problem followed a flash for a different board, see **Recover from wrong firmware** on page 9.

4 Separate the two verification checks

OneROM verifies the flash write. TunerPro verifies the ROM through the running emulator. For a TunerPro mismatch, check the selected plugin, ROM size and BIN, then rebuild with the current app.

Test coverage for this release

Successful hardware testing was reported on Fire 28C and Fire 32B. Fire 28A, 28B and 32A were covered by software checks for this release.

Downloads, release notes and support
RapidFire-001 / OneROM v3 on GitHub

Recover from wrong firmware

A physical Fire 28C flashed with Fire 28A firmware may report 28A or stop responding. Check the revision printed on the PCB before choosing a recovery image.

Check PCB → Connect over USB → Program and verify

1 Connect in One ROM Web

Disconnect the Fire board from its host. Open **One ROM Web** in Edge or Chrome. Use a USB data cable, click **Connect** and select the Fire board or its bootloader entry.

2 Enter BOOTSEL if the board is missing

Unplug USB. Hold **BOOTSEL** to ground while reconnecting USB. Use the official pin guidance below. Remove the bridge once the bootloader appears. Connect to **RP2350 Boot** or **One ROM Bootloader**.

3 Select the exact firmware and program

In **One ROM Builder**, select the physical revision under **One ROM Type**. Choose compatible firmware, add your ROM and matching chip type, then click **Build Firmware** and **Program**.

Read the browser warning before continuing

Board mismatch can compare the old firmware's reported type with your new image. Continue only when the new image matches the physical PCB. If the board type cannot be checked, verify its marking. Cancel if unsure or if the image conflicts with the commissioned board type.

Programming replaces the flash contents. Keep USB connected until verification finishes. After restart, check the revision in **Device Information**.

One ROM Web · **Identify your board** · **Official BOOTSEL guidance**